# In Silico Structural Characterization and Functional Annotation of an Uncharacterized Hydrothermal Vent Metagenomic Protein (GenBank: CUS43520.1 / UniParc: UPI0007B67DD0)

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Study Type: Computational Structural Biology & Biocuration Case Study

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## 1. Executive Summary

Nearly half of all sequenced proteins in public genomic databases derived from extreme environments remain unannotated as "hypothetical" or "uncharacterized" proteins. In this case study, we performed an *in silico* structural and functional investigation of CUS43520.1 (UniParc ID: UPI0007B67DD0; former UniProtKB ID: A0A160TFL8; Gene locus: MGWOODY_Smn2333), a 543-amino-acid protein sequenced from a deep-sea hydrothermal vent metagenome that currently has no predicted functional or domain annotations in NCBI or UniProt.

Using AlphaFold 3 *de novo* structure prediction (`pTM = 0.82`) combined with Foldseek 3D structural alignments across experimental (`PDB100`) and predicted (`AFDB50`, `AFDB-SwissProt`) structure databases, we demonstrate that:

1. CUS43520.1 folds into a rigid, high-confidence bilobal globular architecture containing a well-defined central catalytic cleft.
2. Resolution of Database Misannotation: In related bacterial species, homologs of this protein exist as larger (~1,073 aa) two-domain fusion proteins containing an N-terminal Tetratricopeptide Repeat (TPR) domain (residues 1–530) fused to a C-terminal enzymatic domain (residues 531–1073). Automated annotation pipelines mislabeled many of those full-length proteins simply as *"Tetratricopeptide repeat proteins."* Structural alignment proves that CUS43520.1 lacks the N-terminal TPR region entirely and represents a

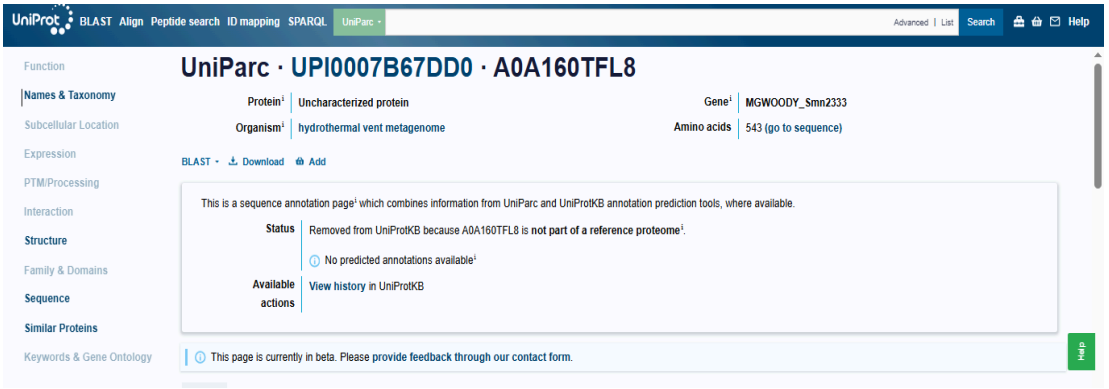
standalone 543-aa C-terminal Restriction Endonuclease-like enzyme (`TM-score = 0.936 – 0.965`).

3. Conserved Active Site & Fold Topology: The internal α/β core (residues 161–289) shares distant structural topology (`TM-score = 0.561`, `Probability = 0.96`) with an experimentally solved metallopeptidase from the deep-sea submarine volcano bacterium *Idiomarina loihiensis* (`PDB: 4FGM`), while the central cleft houses a strictly conserved putative catalytic motif (`IHSWDEARW...ERWRERFG`).

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## 2. Target Information ("Before" Status)

- NCBI GenBank Accession: `CUS43520.1`
- UniProt / UniParc Identifier: `A0A160TFL8` / `UPI0007B67DD0`
- Gene / Sample Locus Tag: `MGWOODY_Smn2333`
- Source Organism: `hydrothermal vent metagenome`
- Sequence Length: 543 amino acids
- Prior Database Annotation: Listed as *"hypothetical protein"* (NCBI) and *"Uncharacterized protein — No predicted annotations available"* (UniParc).




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## 3. Methodology

1. Sequence Retrieval: The 543-residue FASTA sequence (`CUS43520.1`) was retrieved from NCBI GenBank and cross-referenced against UniProtKB/UniParc (`UPI0007B67DD0`).
2. 3D Structure Prediction: The primary amino acid sequence was submitted to the AlphaFold 3 Server. Model quality was evaluated using the predicted Template Modeling score (`pTM`), per-residue confidence (`pLDDT`), and Predicted Aligned Error (`PAE`) matrices.
3. 3D Structural Homology Search: The top-ranked structural model (`model_0.cif`) was queried against Foldseek (`3Di/AA` structural alphabet

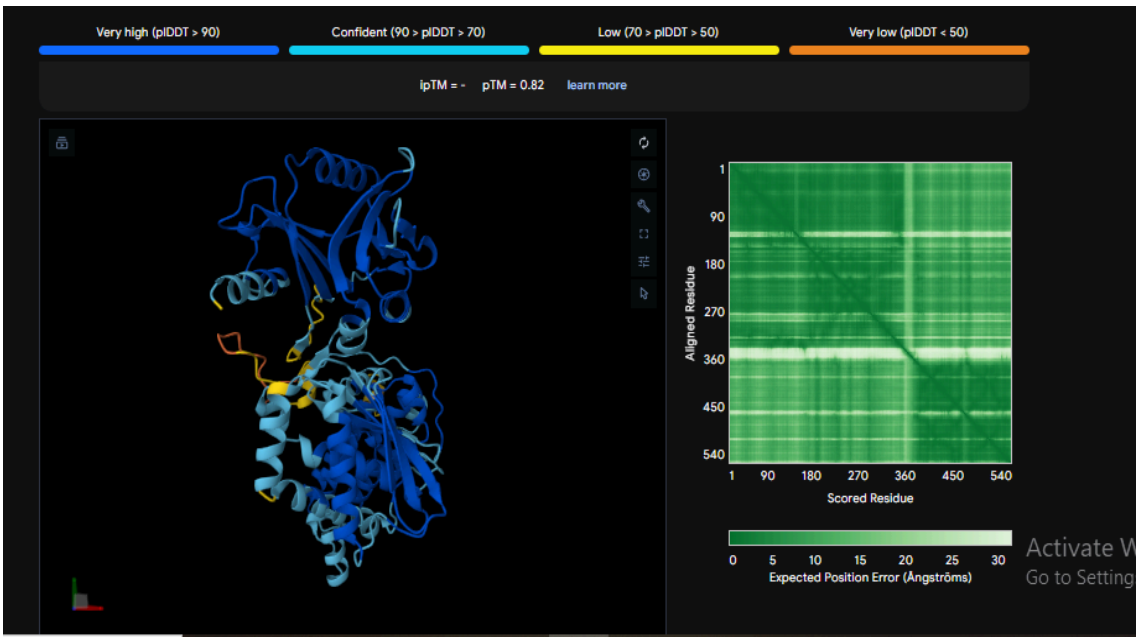
alignment mode) across 10 structural databases, including `PDB100` (experimentally solved structures), `AFDB-SwissProt` (curated proteins), and `AFDB50`. Structural similarity was quantified using `TM-score`, Root-Mean-Square Deviation (`RMSD`), `E-value`, and alignment probability.

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# 4. Results & Discussion

## 4.1. AlphaFold 3 Structure & Confidence

AlphaFold 3 converged on a compact, well-ordered 3D structure with a global `pTM = 0.82`, substantially exceeding the `0.50` threshold for reliable global fold prediction. Aside from a short flexible N-terminal tail (residues 1–15), the protein exhibits high local confidence (`pLDDT > 90`) across two distinct lobes that pack tightly together (as confirmed by low off-diagonal error in the PAE matrix) to form a central substrate-binding pocket.



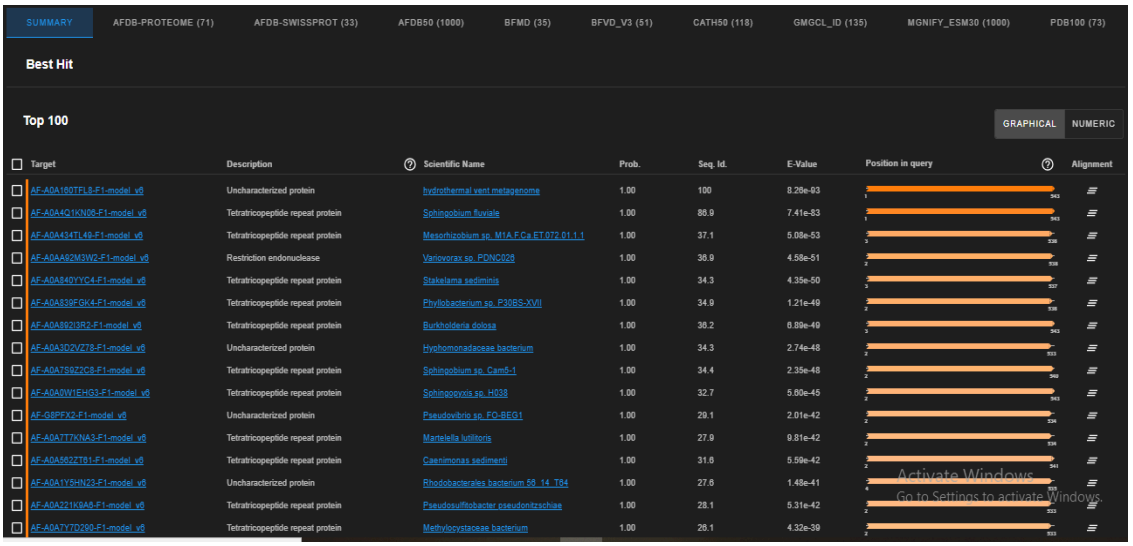

## 4.2. Structural Alignment & Domain Architecture (AFDB50)

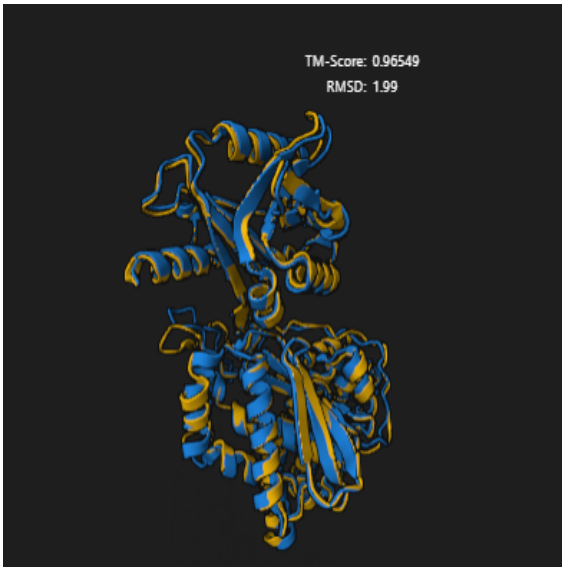
Foldseek search against `AFDB50` identified full-length structural homologs with `Probability = 1.00`:

- Homolog 1 (`AF-A0A4Q1KN06-F1`, *Sphingobium fluviale*): `TM-score = 0.96549`, `RMSD = 1.99 Å`, `Seq. Id. = 86.9%`, `E-value = 7.41e-83`.

- **Homolog 2 (`AF-A0AA92M3W2-F1`, Variovorax sp.):** Annotated as Restriction endonuclease; `TM-score = 0.93592`, `RMSD = 2.30 Å`, `Seq. Id. = 36.9%`, `E-value = 4.58e-51`.

Key Domain Finding: Sequence-structure mapping revealed that residue `1` (`MAAAG...`) of CUS43520.1 aligns starting at residue `531` (`A0A4Q1KN06`) and residue `533` (`A0AA92M3W2`) of its bacterial homologs. In those organisms, residues `1–530` form an N-terminal Tetratricopeptide Repeat (TPR) scaffolding domain, which caused automated annotation pipelines to mislabel the entire 1,073-aa protein as a *"TPR protein."* CUS43520.1 lacks the `1–530` TPR domain completely and consists solely of the 543-aa C-terminal Restriction Endonuclease-like catalytic unit.

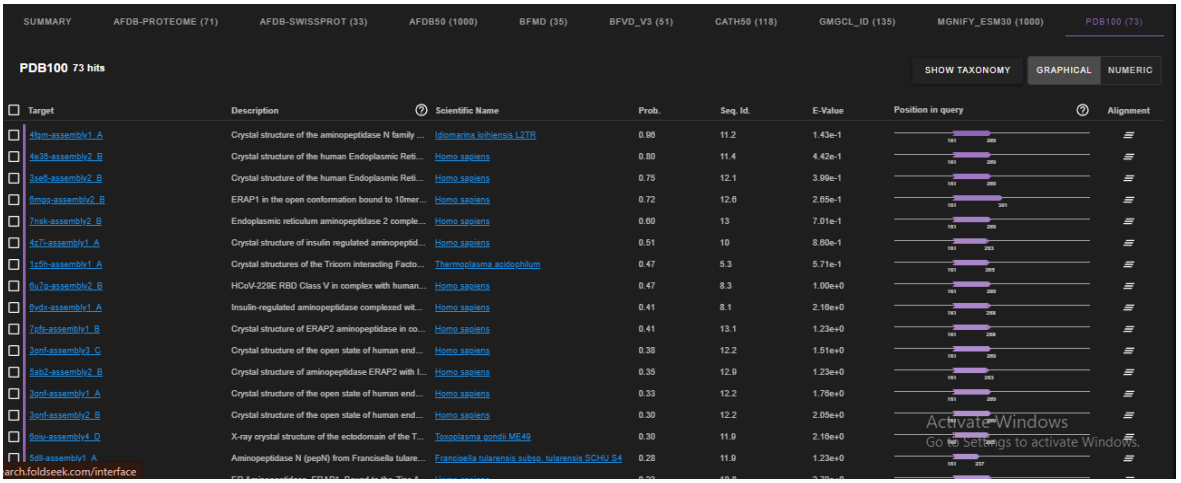





## 4.3. Distant Experimental Fold Match (PDB100)

No full-length crystal structure of this protein family currently exists in the Protein Data Bank (`PDB100`). However, local structural alignment of the central α/β core (residues `161–289`) matched the crystal structure of an M1-family aminopeptidase/metallopeptidase from the deep-sea hydrothermal vent bacterium *Idiomarina loihiensis* L2TR (`PDB: 4FGM`, `Probability = 0.96`, `TM-score = 0.56125`, `RMSD = 4.72 Å`), despite sharing only `11.2%` sequence identity. This confirms a conserved ancestral α/β hydrolase/nuclease core scaffold adapted to hydrothermal environments.






### 4.4. Identification of Putative Catalytic Motif

Despite `63.1%` overall sequence divergence between CUS43520.1 and *Variovorax sp.* (`AF-A0AA92M3W2`), structural alignment identified a highly conserved block spanning residues `365–415` inside the central cleft: `VQSVIDIHSWDEARWKGILYASYGDDVPPIFALTFTNAAGARRIFERWRERFG` The strict conservation of spatially clustered acidic (`D`, `E`) and basic (`H`, `R`, `K`) residues (`IHSWDEARW`, `SYGDD`, and `ERWRERFG`) strongly indicates a metal-coordinating active site responsible for nucleic acid or peptide cleavage.

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## 5. Repository Directory Structure

- `01_Sequence_Data/` — Raw NCBI FASTA sequence (`CUS43520.1.fasta`) and UniParc baseline record (`UPI0007B67DD0`).
- `02_AlphaFold3_Structure/` — Top-ranked 3D coordinate file (`model_0.cif`), confidence metrics (`summary_confidences_0.json`), PAE matrix data (`full_data_0.json`), and structure visualization.
- `03_Foldseek_Alignments/` — Full tabular Foldseek alignment archives (`.m8` tables across all databases), 3D superposition files (`.cif`), and structural alignment renderings (`.png`) for `4FGM`, `A0A4Q1KN06`, and `A0AA92M3W2`.
- `04_Final_Report/` — Complete case study documentation.